(1) Data ware house for City Hospital Consists of the following four dimensions.

AS The three popular schemas used in data ware housing are star schema, snowflake and fact constellation schema.

For the given data ware house, we can use a star schema where Time, Doctor and patient are the dimension tables Connected to a central fact table. The fact table contains the two Measure Count and Charge.

The relational database contains the tables fee (day, month year, doctor, hospital, Patient, count, Charge). The SQL query to retrieve the data is SELECT day, month, year, doctor, hospital, patient, count charge FROM Fee.

(2) A smart city project is collecting data from multiple Sources.

AS A snowflake scheme for big university contains four dimensions: Student Course semester and Instructor with Count and Average Grade as measures in the fact table. To find the average grade of CS Course for each student. we first select the CS Course using.

a slice or dice operation, then roll up the required dimensions and group the result by student to calculate the average grade. if each of the four dimensions have five levels. the total number of cuboides is $5 \times 5 \times 5 \times 5 = 625$ cuboids, including the base and apex cuboids.

3) Weather Data warehouse:-

A) The weather data warehouse is designed to store data collected from about 1000 probs every hour for more then then years. The main dimension are time and location where time contains levels such as hours, day, month and year and location contains probe and region details. The fact table store the measures air pressure, temperature and precipitation OLAP operations such as roll-up, drill down, slice and dices can be used to analyze weather conditions and Identify general weather patterns based on time and location.

(4) Data Cube:

A data cube has n dimension and each dimension has p distinct of cells in the base cuboid in p^n, while the minimum number of cells is 1. since there are no concept hierarchies each dimension has a base level and an All level. Therefore the minimum number of cells in the minimum number of cells in the complete data cube is 2^n.

(5) University Academic performance Data ware house.

AS The University academic performance data ware house uses Student, Course faculty and time as dimension. The central fact table Contains the measure total marks and attendance percentage. To display average marks department wise per each academic year. we roll up the student dimension to department and the Time dimension to year.